Sample Size in the Population Proportion Case.

Q: What is the necessary sample size w/ a required margin of error m?

→:

$$\text{pt. estim.} \pm \text{margin of error}$$

$$\hat{p} \pm \boxed{z^* \cdot \text{std error} \quad z^* \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

w/ z^* the critical value of $N(0,1)$ for the confidence level C

$$z^* \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq m \quad /^2$$

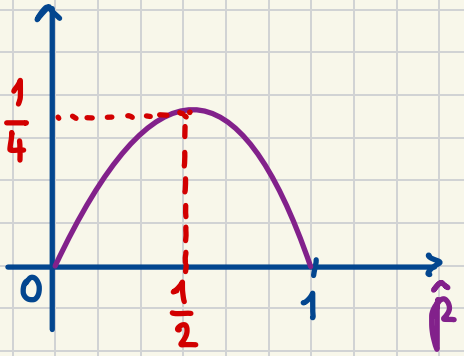
$$(z^*)^2 \cdot \frac{\hat{p}(1-\hat{p})}{n} \leq m^2$$

$$(z^*)^2 \cdot \frac{\hat{p}(1-\hat{p})}{m^2} \leq n$$

We cannot know $\hat{p}$ before knowing n.

Idea #1. Use a $\hat{p}$ from a prior study.

Idea #2. Consider $\hat{p}(1-\hat{p})$ as a f'tion of $\hat{p}$.



The conservative choice for $\hat{p}$ is $1/2$.

$\Rightarrow$ The conservative choice for the sample size.

$$n \geq \frac{(z^*)^2}{4m^2} = \left(\frac{z^*}{2m}\right)^2$$

Confidence Intervals for the Variance.

Consider a normal random sample $(Y_1, \dots, Y_n)$ w/ both parameters μ and σ unknown.

A **good** (unbiased) estimator for σ^2 is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

Theorem. In the above framework, set

$$Q^2 = \frac{1}{\sigma^2} \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

Then,
- $Q^2 \sim \chi^2(df = n-1)$
- Q^2 is a pivotal quantity for σ^2

Problem 16.3. *What is the unbiased estimator for σ^2.*

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

Problem 16.4. *Assume a random sample $Y_1, Y_2, \ldots, Y_n$ from a normal distribution with mean μ and standard deviation σ - both unknown. What's the distribution of*

$$Q^2 = \frac{\sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2}\ ?$$

$Q^2 \sim \chi^2(df = n-1)$

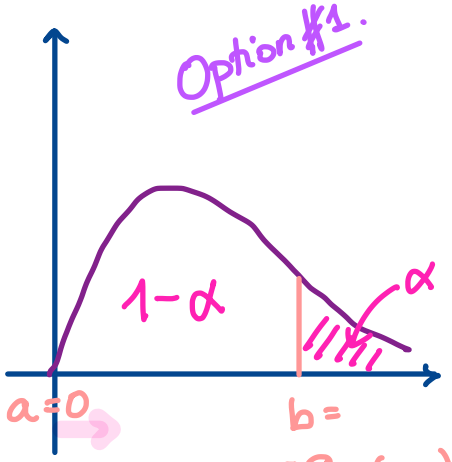
a PIVOTAL QUANTITY for σ^2

Problem 16.5. *Assume that you are assigned a* confidence level $1-\alpha$. *What does it mean to find a* confidence interval *for σ^2?*

$$\mathbb{P}[a \le Q^2 \le b] = 1-\alpha$$

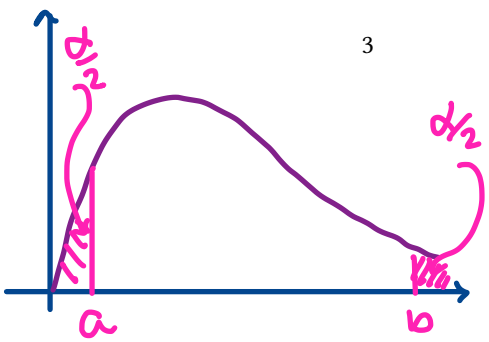
$$\mathbb{P}\left[a \le \frac{(n-1)\cdot S^2}{\sigma^2} \le b\right] = 1-\alpha$$

Option #1.



Problem 16.6. *Are χ^2_1 and χ^2_0 as above uniquely defined?*

No :)



We can choose a symmetric confidence interval

a = qchisq(α/2, df = n-1)

b = qchisq(1-α/2, df = n-1)

Problem 16.7. *What's the form of the confidence interval, then?*

$$\mathbb{P}\left[\frac{1}{a} \geq \frac{\sigma^2}{(n-1)S^2} \geq \frac{1}{b}\right] = 1-\alpha$$

$$\mathbb{P}\left[\frac{(n-1)S^2}{a} \geq \sigma^2 \geq \frac{(n-1)S^2}{b}\right] = 1-\alpha$$

$\hat{\theta}_R$ $\hat{\theta}_L$

Problem 16.8. *Assume the above setting. Let the random sample be of size $n = 9$. You do the arithmetic and arrive at the estimate $s^2 = 7.93$ (based on the data set). Using the above procedure, find the $90\%-$confidence interval for σ^2.*

→ : $n=9$

$s^2=7.93$

$\alpha = 0.10$

Our confidence interval:

$$\left(\frac{8 \cdot 7.93}{b}, \frac{8 \cdot 7.93}{a}\right) = (4.091, 23.216)$$

$b = 15.51$ $a = 2.733$

$a = \text{qchisq}(0.05, df = 8)$

$b = \text{qchisq}(0.95, df = 8)$

□

Confidence Intervals for the μ w/ Variance Unknown.

Focus on the normal model $N(\mu, \sigma)$ w/ both parameters unknow, but, w/ the target parameter μ.

Theorem.

- $\bar{Y} \sim \text{Normal}\left(\text{mean} = \mu, \text{sd} = \frac{\sigma}{\sqrt{n}}\right)$ ←

$$\bar{Y} = \frac{1}{n}\underbrace{(Y_1 + \cdots + Y_n)}_{\text{Normal}}$$

- $Q^2 = \frac{1}{\sigma^2}\sum_{i=1}^{n}(Y_i - \bar{Y})^2 \sim \chi^2(df = n-1)$ ←
- $\bar{Y}$ and Q^2 are independent.

Goal: Confidence interval for μ.

Idea:

$$U = \frac{\bar{Y} - \mu}{\frac{S}{\sqrt{n}}}$$

w/

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

pivotal quantity?

$$\frac{\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}}{\frac{S}{\sigma}} = \frac{Z}{\sqrt{\frac{S^2}{\sigma^2}}} = \frac{Z}{\sqrt{\frac{Q^2}{n-1}}}$$

$\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$